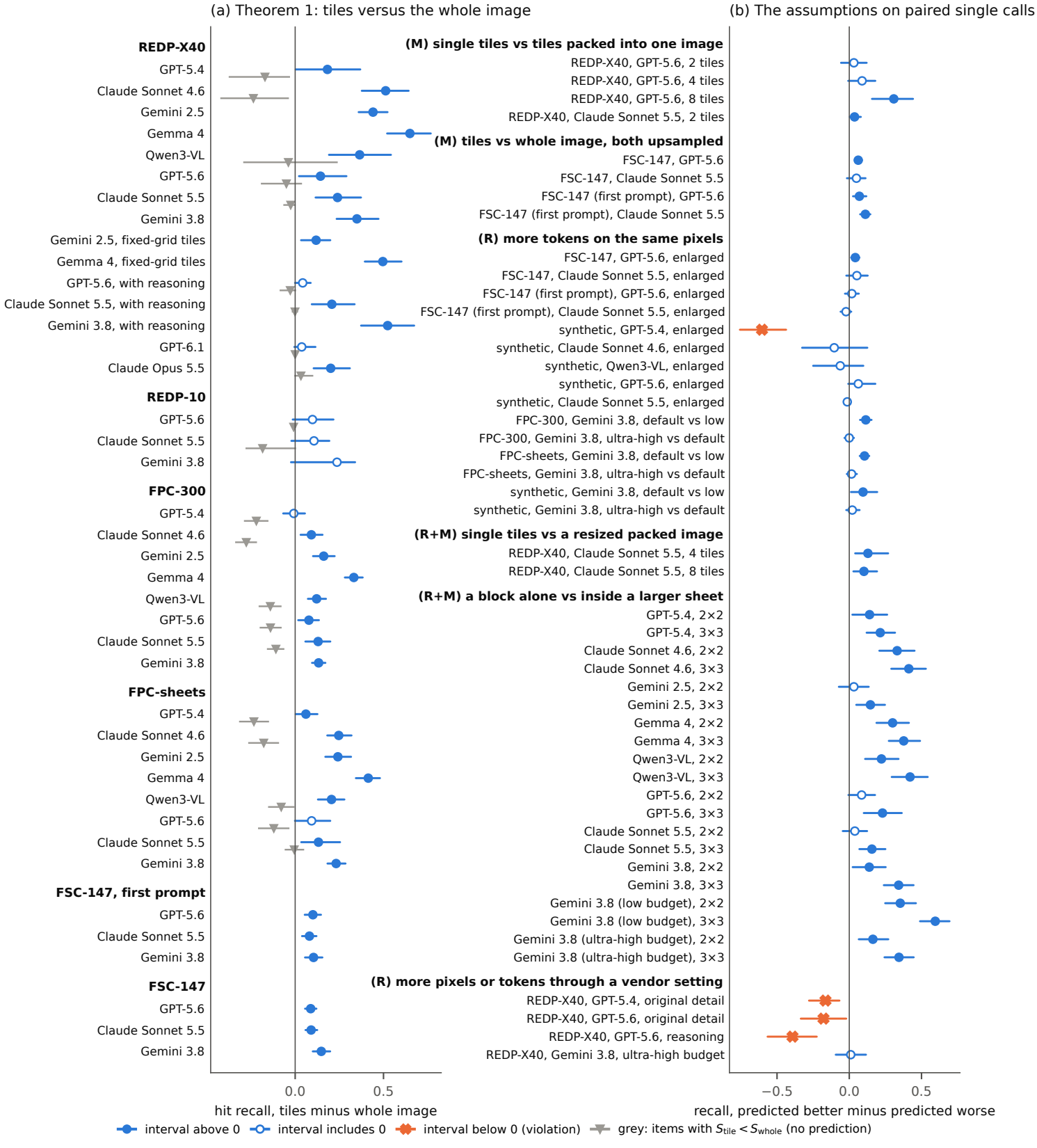
(a) Theorem 1: tiles versus the whole image
(b) The assumptions on paired single calls
REDP-X40
GPT-5.4
Claude Sonnet 4.6
Gemini 2.5
Gemma 4
Qwen3-VL
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
Gemini 2.5, fixed-grid tiles
Gemma 4, fixed-grid tiles
GPT-5.6, with reasoning
Claude Sonnet 5.5, with reasoning
Gemini 3.8, with reasoning
GPT-6.1
Claude Opus 5.5
REDP-10
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
FPC-300
GPT-5.4
Claude Sonnet 4.6
Gemini 2.5
Gemma 4
Qwen3-VL
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
FPC-sheets
GPT-5.4
Claude Sonnet 4.6
Gemini 2.5
Gemma 4
Qwen3-VL
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
FSC-147, first prompt
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
FSC-147
GPT-5.6
Claude Sonnet 5.5
Gemini 3.8
0.0
0.5
hit recall, tiles minus whole image
(M) single tiles vs tiles packed into one image
REDP-X40, GPT-5.6, 2 tiles
REDP-X40, GPT-5.6, 4 tiles
REDP-X40, GPT-5.6, 8 tiles
REDP-X40, Claude Sonnet 5.5, 2 tiles
(M) tiles vs whole image, both upsampled
FSC-147, GPT-5.6
FSC-147, Claude Sonnet 5.5
FSC-147 (first prompt), GPT-5.6
FSC-147 (first prompt), Claude Sonnet 5.5
(R) more tokens on the same pixels
FSC-147, GPT-5.6, enlarged
FSC-147, Claude Sonnet 5.5, enlarged
FSC-147 (first prompt), GPT-5.6, enlarged
FSC-147 (first prompt), Claude Sonnet 5.5, enlarged
synthetic, GPT-5.4, enlarged
synthetic, Claude Sonnet 4.6, enlarged
synthetic, Qwen3-VL, enlarged
synthetic, GPT-5.6, enlarged
synthetic, Claude Sonnet 5.5, enlarged
FPC-300, Gemini 3.8, default vs low
FPC-300, Gemini 3.8, ultra-high vs default
FPC-sheets, Gemini 3.8, default vs low
FPC-sheets, Gemini 3.8, ultra-high vs default
synthetic, Gemini 3.8, default vs low
synthetic, Gemini 3.8, ultra-high vs default
(R+M) single tiles vs a resized packed image
REDP-X40, Claude Sonnet 5.5, 4 tiles
REDP-X40, Claude Sonnet 5.5, 8 tiles
(R+M) a block alone vs inside a larger sheet
GPT-5.4, 2×2
GPT-5.4, 3×3
Claude Sonnet 4.6, 2×2
Claude Sonnet 4.6, 3×3
Gemini 2.5, 2×2
Gemini 2.5, 3×3
Gemma 4, 2×2
Gemma 4, 3×3
Qwen3-VL, 2×2
Qwen3-VL, 3×3
GPT-5.6, 2×2
GPT-5.6, 3×3
Claude Sonnet 5.5, 2×2
Claude Sonnet 5.5, 3×3
Gemini 3.8, 2×2
Gemini 3.8, 3×3
Gemini 3.8 (low budget), 2×2
Gemini 3.8 (low budget), 3×3
Gemini 3.8 (ultra-high budget), 2×2
Gemini 3.8 (ultra-high budget), 3×3
(R) more pixels or tokens through a vendor setting
REDP-X40, GPT-5.4, original detail
REDP-X40, GPT-5.6, original detail
REDP-X40, GPT-5.6, reasoning
REDP-X40, Gemini 3.8, ultra-high budget
−0.5
0.0
0.5
recall, predicted better minus predicted worse
interval above 0
interval includes 0
interval below 0 (violation)
grey: items with $S_{\text{tile}} < S_{\text{whole}}$ (no prediction)